

Report
Of
Financial Cybersecurity Web Platform & Secure Cloud Architecture

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B. Tech CSE



By
Roll Number of Student: 25SCS1003004870
Name of Student: SHUBHAM CHAUDHARY
Batch: 2CSE27

2026-27

1. CANDIDATE'S DECLARATION

I, Shubham Chaudhary, do hereby declare that the internship report titled "Financial Cybersecurity Web Platform & Secure Cloud Architecture" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: *Shubham*

Student Name: Shubham Chaudhary

Roll No.: 25SCS1003004870

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents, Mr. Ram Ratan and Mrs. Madhubala, for their constant support and blessings, and to the faculty of the School of Computer Science and Engineering, IILM University, for their guidance throughout this internship.

Signature: *Shubham*

Student Name: Shubham Chaudhary

Roll No.: 25SCS1003004870

3. Table of Contents

Sr. No.	Chapter Name	Page No.
1	Candidate's Declaration	i
2	Acknowledgement	ii
3	Table of Contents	iii
4	Internship Completion Certificate	iv
5	Project Description	
	5.1 Introduction	1
	5.2 Organization Profile	1
	5.3 Problem Statement	1
	5.4 Project Objectives	2
	5.5 Scope of the Project	2
	5.6 Technologies and Tools Used	2
	5.7 System Architecture	3
	5.8 Methodology	4
	5.9 Expected Outcomes	5
	5.10 Certificate of Completion	5
6	Bibliography / References	6

4. Internship Completion Certificate



Inspiring and
empowering
future professionals

Shubham Chaudhary **Introduction to Software Engineering Job** **Simulation**

Certificate of Completion
August 23rd, 2026

Over the period of August 2026, Shubham Chaudhary has completed practical tasks in:

Create a Website
Financial Cybersecurity
Stylize your Website
Write a Web Hosting Proposal

A handwritten signature in black ink, appearing to read "Tom Brunskill".

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a6aee7ceef2f057ed31c38 | User Verification Code 6a6aee0dc838adee5c9d423e | Issued by Forage

5. Project Description

5.1 Introduction

This report documents the engineering tasks, methodologies, and technical deliverables completed during the Commonwealth Bank Software Engineering Virtual Internship, offered through the Forage platform. The primary objective of the internship was to architect, style, and outline a secure web-based digital client education platform focused on financial cybersecurity. The project simulated real-world enterprise software engineering workflows at a premier banking institution, exposing the intern to the practices followed by professional software engineering teams.

Throughout the internship, four core milestones were executed: foundational markup structuring using HTML5, client-centric cybersecurity content synthesis, enterprise brand alignment using CSS3 in accordance with strict Commonwealth Bank styling standards, and the formulation of a comprehensive cloud hosting and infrastructure security proposal compliant with international banking regulations (APRA CPS 234).

5.2 Organization Profile

Commonwealth Bank of Australia (CommBank) is one of the leading integrated financial services organizations in the Asia-Pacific region, providing retail, business, and institutional banking services to millions of customers. As digital banking adoption continues to accelerate, the intersection of user experience, software engineering, and proactive cybersecurity has become pivotal to the organization's operations.

The Software Engineering Virtual Program, hosted on the Forage platform, exposes candidates to real-world engineering problem statements that simulate the duties of an entry-level software engineer at CommBank. Engineers on this program are required to balance technical robustness with accessible communication and enterprise design consistency in order to serve the bank's retail and commercial clients effectively.

5.3 Problem Statement

Retail banking customers are increasingly exposed to digital threats such as credential stuffing, SIM swapping, phishing, and man-in-the-middle attacks over unsecured Wi-Fi networks. Commonwealth Bank required a lightweight, brand-compliant web portal that could educate customers on these risks in clear, non-technical language, while also

needing a hardened cloud hosting architecture capable of securely and reliably delivering this content at enterprise scale. The challenge, therefore, was twofold: building an accessible front-end client education platform, and designing a secure, compliant infrastructure to host it.

5.4 Project Objectives

- Engineer a fully accessible, semantic HTML5 client security portal.
- Author actionable, non-technical cybersecurity guidance across five critical attack vectors: Authentication, Multi-Factor Authentication (MFA), Phishing Defense, Network Encryption, and Device Hygiene.
- Implement responsive, strict brand-compliant typography and visual hierarchy using the corporate colour palette.
- Formulate a hardened enterprise hosting architecture incorporating TLS 1.3, a Layer 7 Web Application Firewall (WAF), DDoS mitigation, and Zero-Trust access protocols.

5.5 Scope of the Project

The scope of this internship project was limited to the design and conceptual development of a static, informational client-education web portal, along with an accompanying hosting and security proposal. It covered front-end markup and styling, cybersecurity content authoring, and cloud infrastructure planning. The scope did not extend to back-end development, live deployment, or integration with CommBank's actual production banking systems, as the exercise was a simulated job task rather than a production engagement.

5.6 Technologies and Tools Used

- HTML5 — for semantic, accessible page structure.
- CSS3 — for responsive, brand-compliant styling.
- Visual Studio Code — as the local development environment.

- AWS CloudFront / Cloudflare CDN — for content delivery and edge protection.
- AWS S3 — for immutable static asset storage.
- TLS 1.3, Let's Encrypt / AWS ACM — for encrypted transport and certificate management.
- GitHub Actions — for CI/CD pipeline automation, including SAST and dependency scanning (Dependabot / Snyk).
- AWS CloudTrail — for immutable audit logging and compliance monitoring.

5.7 System Architecture

While the client education portal itself consists of static informational assets, enterprise financial systems must be designed for secure scalability. The proposed hosting architecture follows a defense-in-depth topology, comprising the following layers:

- Edge Protection & Content Delivery: A global CDN (AWS CloudFront / Cloudflare), coupled with a Web Application Firewall (WAF) to inspect Layer 7 HTTP/HTTPS traffic and absorb Layer 3/4 volumetric DDoS attacks.
- Encrypted Transport: Strict TLS 1.3 enforcement with automated certificate renewal via Let's Encrypt / AWS ACM, and HTTP Strict Transport Security (HSTS) preloading to preclude SSL stripping.
- Immutable Storage & Serverless Delivery: Static assets hosted in an isolated AWS S3 bucket with public access blocked, delivered strictly through CloudFront Origin Access Control (OAC).
- CI/CD & DevSecOps Automation: An automated deployment pipeline via GitHub Actions, integrating static application security testing (SAST) and automated dependency vulnerability checks.

5.8 Methodology

The project was executed across four sequential sprints, each representing a key phase of front-end development, content architecture, brand integration, and cloud operations.

- Sprint 1 — Environment & Markup Setup: Established the foundational workspace in Visual Studio Code and created modular file structures (index.html, index.css) with normalized box-sizing models to ensure consistent rendering across browsers.
- Sprint 2 — Content Integration: Conducted domain research into contemporary threat vectors facing retail banking clients, including credential stuffing, SIM swapping, phishing, and man-in-the-middle exploits over unsecured Wi-Fi, and translated these into roughly 350 words of structured, semantic copy.
- Sprint 3 — Brand Styling Compliance: Applied CommBank's corporate brand guidelines, including the Primary Yellow (#FFCC00) and Beacon Yellow (#FFFF00) colour palette, a unified sans-serif typography system, and strict proportional scaling for headings and body text.
- Sprint 4 — Web Hosting Proposal: Devised a multi-tier cloud hosting architecture emphasizing data integrity, encryption, and DDoS defense, culminating in a technical proposal addressing TLS 1.3, WAF, CI/CD security, and APRA compliance.

The client education content itself addressed five core attack surfaces: credential hardening through unique, multi-word passphrases; multi-factor authentication using hardware keys or TOTP apps in place of SMS verification; phishing and social engineering defense; transport layer awareness on public Wi-Fi networks; and endpoint hygiene through automated patching and real-time alerting.

5.9 Expected Outcomes

The Commonwealth Bank Software Engineering Virtual Internship provided rigorous exposure to enterprise-standard engineering practices. Beyond syntax and styling, the experience underscored the critical responsibility software engineers hold in safeguarding digital trust within financial ecosystems. Key competencies acquired include:

- **Front-End Engineering:** Mastery of semantic HTML5 document structuring and CSS3 modular styling adhering to strict corporate design tokens.
- **Technical Communication:** The ability to translate technical security protocols into accessible, empowering guidance for non-technical retail banking clients.
- **Cloud Infrastructure Design:** Architectural competency in designing highly available, hardened cloud hosting environments built on zero-trust principles.
- **Regulatory Governance:** Awareness of banking sector compliance frameworks (APRA CPS 234) and enterprise security baselines.

5.10 Certificate of Completion

The internship was successfully completed on August 23rd, 2026, and a Certificate of Completion was issued by Forage on behalf of Commonwealth Bank, verifying the completion of practical tasks across four modules: Create a Website, Financial Cybersecurity, Stylize your Website, and Write a Web Hosting Proposal. The certificate is reproduced in Chapter 3 of this report (Enrolment Verification Code: 6a6aee7ceef2f057ed31c38; User Verification Code: 6a6aee0dc838adee5c9d423e).



Inspiring and
empowering
future professionals

Shubham Chaudhary

Introduction to Software Engineering Job Simulation

Certificate of Completion
August 23rd, 2026

Over the period of August 2026, Shubham Chaudhary has completed practical tasks in:

Create a Website
Financial Cybersecurity
Styleize your Website
Write a Web Hosting Proposal



Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a6aee7ceef2f057ed31c38 | User Verification Code 6a6aee0dc838adee5c9d423e | Issued by Forage

6. Bibliography / References

1. Forage. Commonwealth Bank Software Engineering Virtual Experience Program. Available at: <https://www.theforage.com/>
2. Australian Prudential Regulation Authority (APRA). CPS 234 — Information Security Prudential Standard.
3. Amazon Web Services. AWS CloudFront, AWS S3, and AWS Certificate Manager Documentation. Available at: <https://docs.aws.amazon.com/>
4. Mozilla Developer Network (MDN). HTML5 and CSS3 Documentation. Available at: <https://developer.mozilla.org/>
5. GitHub. GitHub Actions Documentation. Available at: <https://docs.github.com/actions>